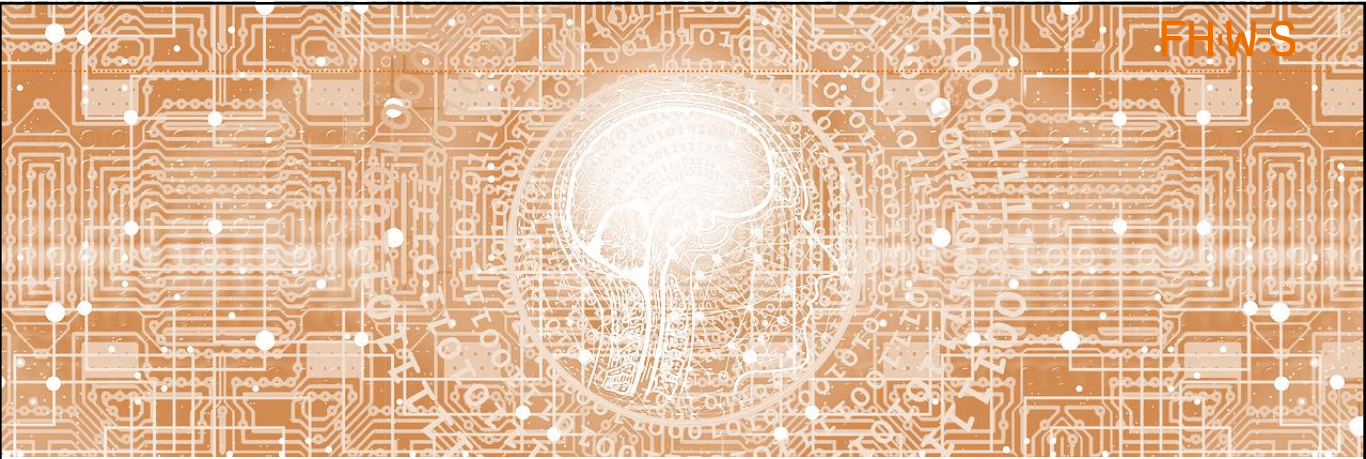




Software Engineering and Cyber Security

Prof. Dr. Marian Daun

FHWS University of Applied Sciences
Würzburg-Schweinfurt



Agenda

- Software Engineering Fundamentals**
- Requirements Engineering**
- Architecture Design**
- Modelling**
- Development and Implementation**
- Software Quality Assurance**

FHWS University of Applied Sciences
Würzburg-Schweinfurt

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Requirements Engineering



Group Discussion

What is Requirements Engineering?

Terminology



Group Discussion

What is a Requirement?

What the IREB says

D

A Requirement is

- (1) **A need perceived by a stakeholder.**
- (2) **A capability or property that a system shall have.**
- (3) **A documented representation of a need, capability or property.**

[IREB Glossary]

**The IREB is the
International Requirements
Engineering Board**

www.ireb.org

D

The goal of RE is to specify and manage requirements for systems such that the systems implemented and deployed satisfy their stakeholders' desires and needs.

[IREB Syllabus CPRE]

Three Types of Requirements

Functional Requirements

Quality Requirements

Constraints

Often Quality Requirements are referred to as Non-functional Requirements.

**However, the term is not good.
Can you guess why?**



Group Discussion

Why do we differentiate between different Types of Requirements?



Group Discussion

The system shall be secure

What is this for a requirement?



Exercise

For your Case System define:

- 10 Functional Requirements**
- 10 Quality Requirements**
- 5 Constraints**

A further Classification



System Requirements



Stakeholder Requirements



User Requirements



Domain Requirements



Business Requirements

Why we need Requirements Engineering

No Silver Bullet

“The hardest single part of building a software system is deciding precisely what to build.

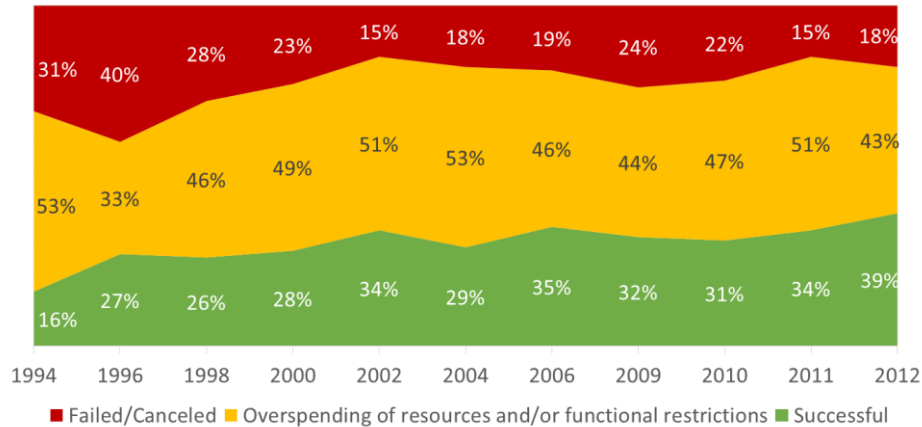
No other part of the conceptual work is as difficult as establishing the detailed technical requirements, including all the interfaces to people, to machines, and to other software systems.

No other part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify later.”

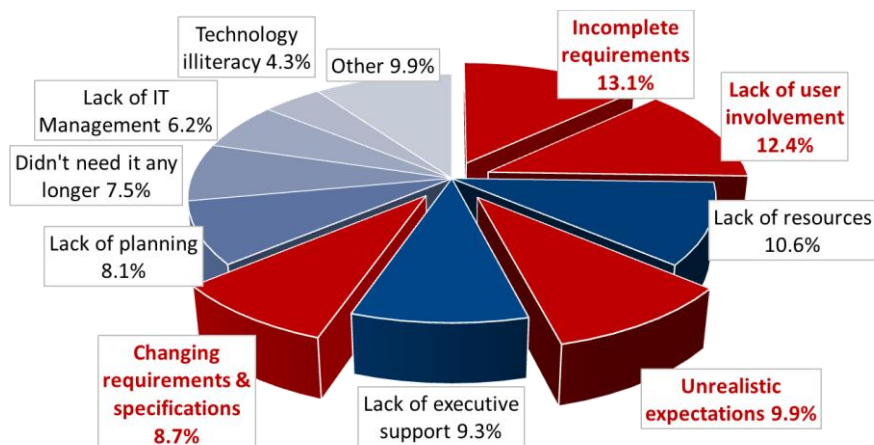
[Brooks 1987]

CHAOS Report

In average only **30%** of the projects were successful, **23%** were cancelled, and **47%** were completed with resource overspend and functional restrictions.



CHAOS Report – Why Projects Fail



Based on [The Standish Group 1995]

What a Requirements Engineer does

Requirements Engineering Activities



**Eliciting
Requirements**



**Documenting
Requirements**

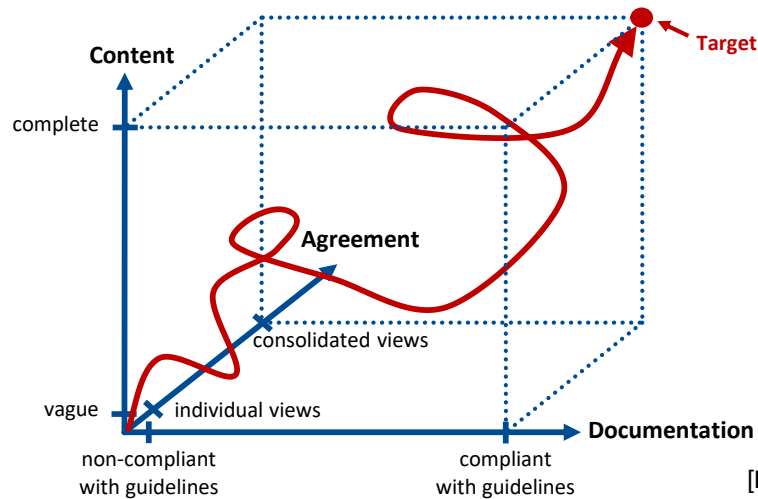


**Validating
Requirements**



**Managing
Requirements**

Three Dimensions of Requirements Engineering



[Pohl and Ulfat-Bunyadi 2013]



Group Discussion

Who is responsible for Requirements Engineering?

Documentation



Group Discussion

Why do we document Requirements?

Importance of Documentation (1)

- **Persistence**

During a project, a **large amount of different types of information** are elicited and developed. Without proper documentation (part of) this information would be forgotten. Moreover, the stakeholders involved can hardly recall and reflect on all this information.

- **Common reference**

Documentation of the relevant information establishes a common reference for all stakeholders.

- **Promotes communication**

The common reference supports communication among the stakeholders and promotes discussions.

- **Promotes objectivity**

Unlike the verbal exchange of information, the exchange of documented information is generally less amenable to unwanted alteration of the information, e.g. due to subjective interpretation and subjective bias in the communication of the information.

Importance of Documentation (2)

- **Support training of new employees**

If a new stakeholder joins the project, the documented information provides an excellent basis for getting familiar with the project. For example, the stakeholder can selectively access the relevant information needed for performing a specific task or can get an overall overview of the system.

- **Preserve expert knowledge**

Typically, not all project members know all relevant aspects of a system such as all relevant technologies or all relevant context aspects. Documenting such expert knowledge makes this knowledge available to all project participants and hence reduces the dependence on individual experts.

- **Supports problem reflection**

When documenting information, the author is forced to structure the information in an appropriate way. Hence the author has to reflect on the information which often leads to the identification of gaps and inconsistencies

Requirements Specification Documents

Representation Formats

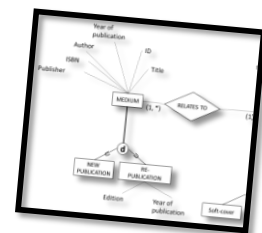
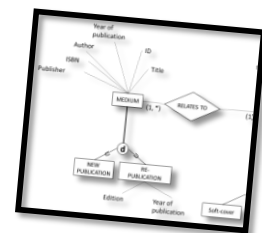
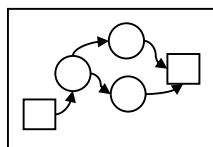
Textual

R-1.1	_____

R-1.2	_____

- Natural language text
- Structured text
- Templates

Model-based



Combined

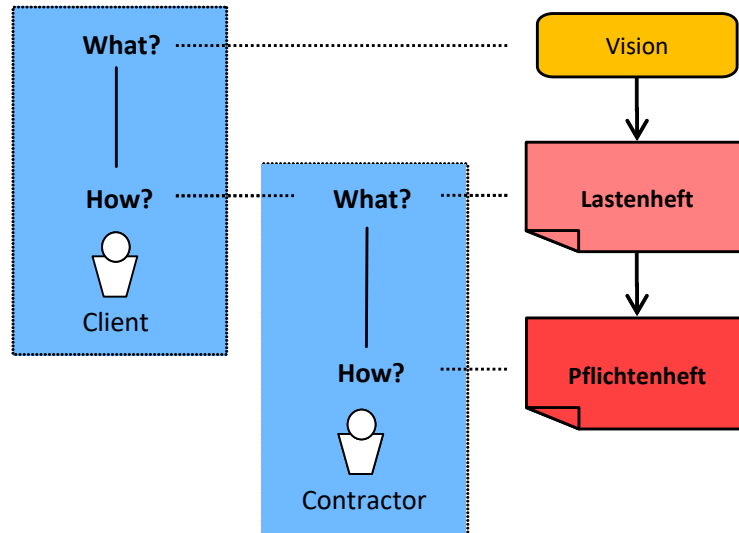
	R-1.1	_____

	R-1.2	_____

- Conceptual models with annotations
- Text with models

Identifier	R-5-33-8
Name	Recommendation for rental car after booking
Author	Matthias Gerdes
Version	V3
Sources	Ms. Reilig, Ms. Meier, TBD (incomplete)
Short description	After booking a journey the system recommends a rental car provided by InterCars.
Requirement	After booking a flight or holiday package the booking system shall display a selection of available rental cars provided by InterCars. The rental cars shown shall be available at the destination point and during the time of the journey booked. Each rental car shall be described by displaying category, type, price per day and picture.

Lastenheft und Pflichtenheft



Lastenheft und Pflichtenheft

Abstract Requirement in the Lastenheft:

R-L-12: The safety system shall offer user and access management.

Detailed Requirements in the Pflichtenheft:

R-8: The safety system shall differentiate between the administrator and user of the system.

...

R-14: The administrator shall be able to create new user accounts.

R-15: The system shall support the organization of user accounts in user groups.

R-16: The administrator shall be able to assign and revoke access authorizations.

...

R-23: The user shall be able to change his/her personal access code.

...

Lastenheft

D The Lastenheft contains a definition of the system vision and a description of the essential system goals (functions and qualities). It names important context objects (e.g. constraints) of the requirements engineering context, as well as their relationships to the vision and the defined system goals.

- Typically created by the client.
- Contains abstract descriptions of the desired functionalities and qualities the system should provide.

Pflichtenheft

D The Pflichtenheft details the vision and the system goals (abstract functionalities and qualities) described in the Lastenheft. If required, it also details the constraints defined in the Lastenheft with regard to the intended technical realization of the system.

- Typically created by the contractor.
- Defines all detailed requirements.
- If required, additional requirements (requirements not stated in the Lastenheft) are added during the creation of the Pflichtenheft.
- Often a Pflichtenheft also specifies the assignment of requirements to an initial architecture



Exercise

For next week create a Requirements Specification for your Case System



Group Discussion

When is a Requirement a good Requirement?

Quality Criteria for Requirements (1)

Agreed: The relevant stakeholders established a sufficient agreement about the requirement – cf. agreement dimension of requirements engineering, Lecture “Requirements Engineering Framework”). based on [Pohl and Rupp 2011]

Atomic: The requirement describes only a single, coherent fact. based on [ISO/IEC/IEEE 29148]

Complete: The requirement adheres to the defined documentation and specification guidelines, does not omit any information relevant for some stakeholder, and does not need any further explanations. based on [ISO/IEC/IEEE 29148]

Comprehensible: The content of the requirement is easy to understand for all stakeholders. based on [Pohl and Rupp 2011]

Consistent: A requirement is consistent if the statements within the requirement do not contradict each other. In addition, the consistency of an entire set of requirements (document) as to be taken into account – cf. quality criteria for requirements specifications. based on [ISO/IEC/IEEE 29148; Pohl and Rupp 2011]

Correct: The relevant stakeholders confirm that the specified or documented requirement is correct, and that the system shall realise the requirement. based on [IEEE Std 830-1998]

Feasible: The requirement can be technically realised within the system constraints, and does not require major technological advancements. based on [ISO/IEC/IEEE 29148]

Quality Criteria for Requirements (2)

Identifiable: The requirement can be uniquely distinguished from other requirements (e.g. by using a unique identifier).

Implementation free: The requirement does not contain any unnecessary constraints regarding the architecture and its realization. based on [ISO/IEC/IEEE 29148]

Necessary: The requirement defines a crucial system property. If it is missing, defects will occur that cannot be resolved by other capabilities. based on [ISO/IEC/IEEE 29148]

Rated: The relevance, priority and stability of the requirement have been defined. based on [Pohl and Rupp 2011]

Traceable: The source, evolution and use in later development phases of the requirement are identified and documented. based on [ISO/IEC/IEEE 29148]

Unambiguous: The requirement only permits one valid interpretation. All readers arrive at the same understanding of the requirement. based on [ISO/IEC/IEEE 29148]

Verifiable: The requirement allows proving whether the implemented system satisfies the requirement or not. based on [ISO/IEC/IEEE 29148]

Quality Criteria for Specifications (1)

Affordable: There exists a feasible solution that satisfies the complete set of requirements within the lifecycle constraints (e.g. cost, time, legal). based on [ISO/IEC/IEEE 29148]

Bounded: The set of requirements is defined within the system scope, i.e. the requirements specification contains no "gold plating". based on [ISO/IEC/IEEE 29148]

Clearly structured (comprehensible): The requirements specification supports selective reading and should thus be clearly structured and comprehensible. based on [Pohl and Rupp 2011]

Complete: The requirements specification contains all relevant requirements, and each requirement is completely specified. No TBDs etc. are included. based on [ISO/IEC/IEEE 29148, IEEE 830-1998]

Quality Criteria for Specifications (2)

Consistent: All single requirements are internally consistently defined. There are no contradictions between the requirements. based on [ISO/IEC/IEEE 29148]

Modifiable and extendable: The requirements specification is easy to modify and extend. The document structure and style should support simple, complete and consistent modification.
based on [Pohl and Rupp 2011, IEEE 830-1998]

Traceable: The relationships between different versions of requirements specifications and other documents (e.g. business process models, architecture, test cases, user manuals, standards) should be traceable.
based on [Pohl and Rupp 2011]

Unambiguous: Each single requirement on its own, as well as the entire requirements specification allow only one valid interpretation. based on [Pohl and Rupp 2011]

Writing good requirements

Avoid ambiguousness

Avoid vagueness

Requirements Sources

Three Types of Requirements Sources

Stakeholder

Documents

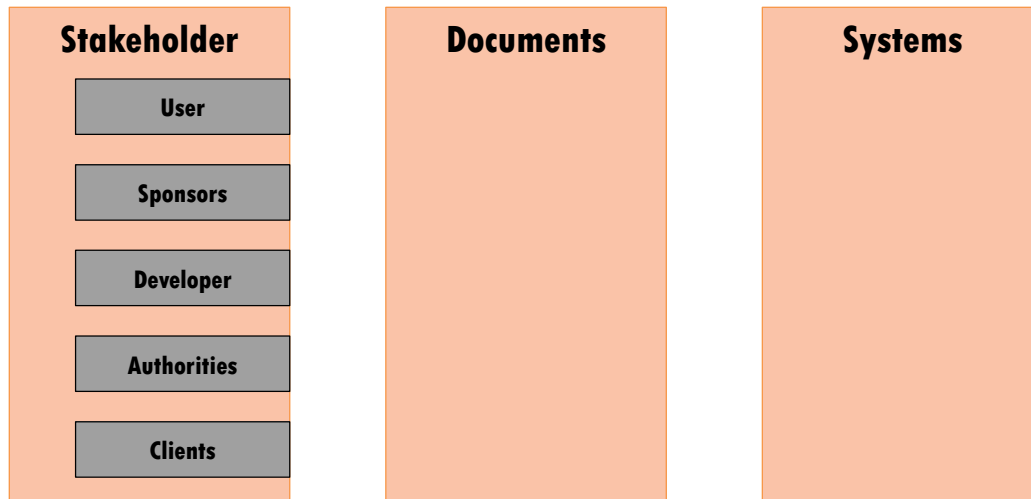
Systems



Group Discussion

Who are relevant Stakeholders?

Three Types of Requirements Sources



Group Discussion

Why are Stakeholders „difficult“?

Manage Stakeholders

Stakeholder are Humans!

Interests

Conflicts

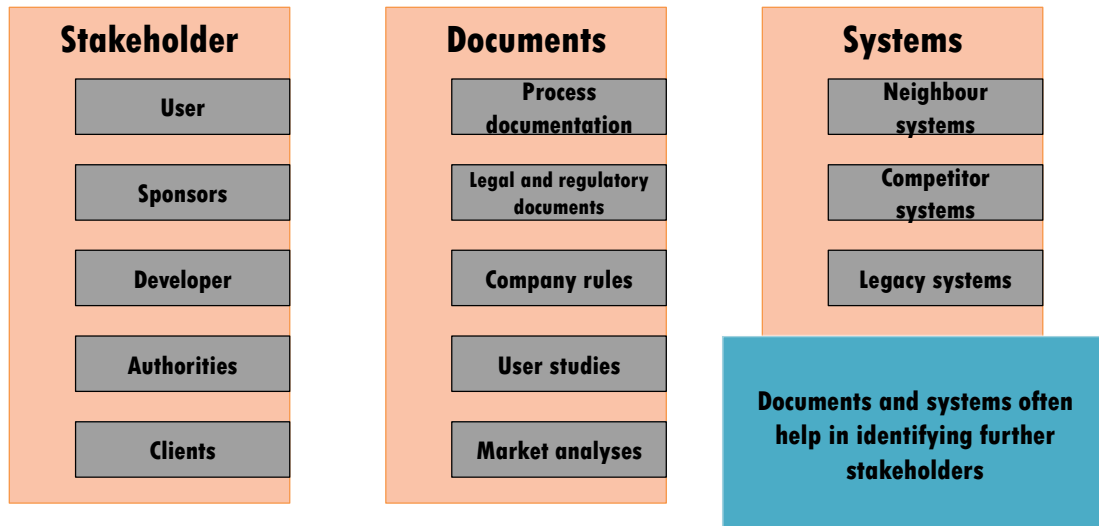
Availability



Group Discussion

What important information about stakeholders should the requirements engineer document?

Drei Arten von Anforderungsquellen

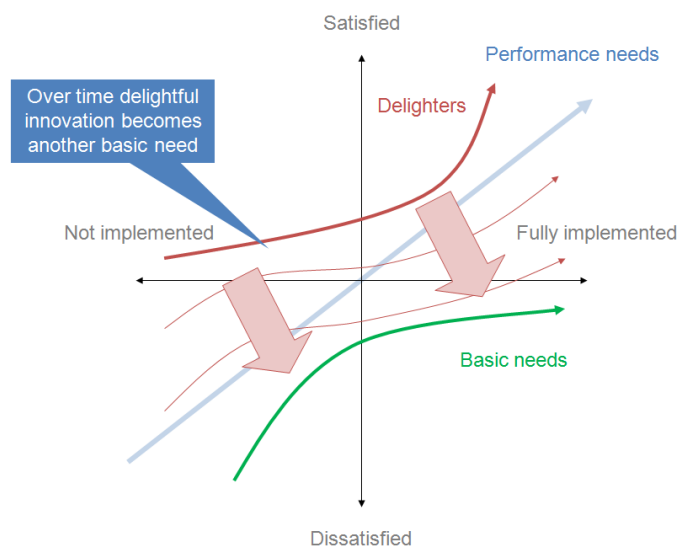


Exercise

Collect Requirements Sources for your Case System

Eliciting Requirements from Requirements Sources

The KANO Model



[Kano et al. 1984, Graphics Wikipedia]



Group Discussion

What Elicitation Techniques exist?

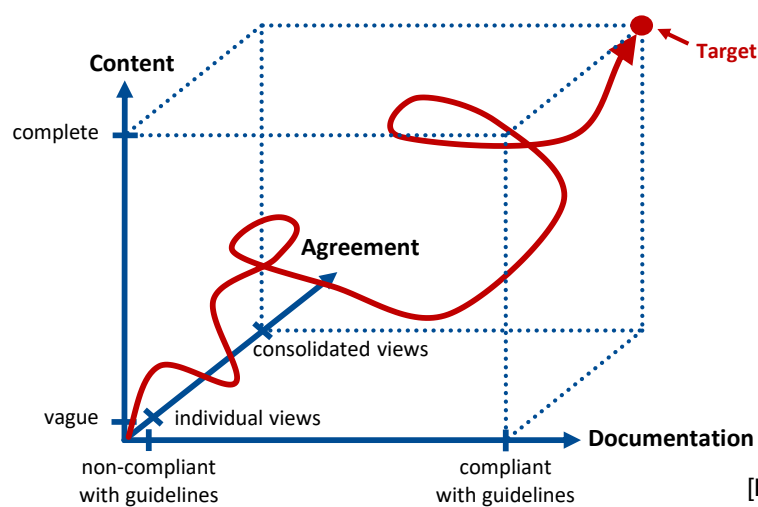


Exercise

Choose appropriate Elicitation Techniques for your Case System

Negotiating Requirements

Three Dimensions of Requirements Engineering



[Pohl and Ulfat-Bunyadi 2013]



Group Discussion

How do Conflicts arise?

Six Conflict Types

**Subject matter
conflict**

Interest conflict

**Relationship
conflict**

Data conflict

Value conflict

**Structural
conflict**

[Moore 2014]



Group Discussion

How can we solve Conflicts?

Techniques for Conflict Resolution

Agreement

Compromise

Voting

Overruling

**Definition of
Variants**

Validating Requirements



Group Discussion

What is the Difference between Validation and Verification?

**Question is also very important
for Software Quality Assurance**

Important Principles

Involvement of the right stakeholders

Separating the identification and the correction of defects

Validation from different views

Repeated Validation

[IREB Syllabus CPRE]

Commonly used validation techniques

Walkthrough

**Artifact
commentary
(Stellungnahme)**

Inspection

Prototyping

**Empirical
evaluation**

**Sample
development**

Managing Requirements

Overview



Questions for Self-Assessment

- What is a requirement?
- Why is it important to differentiate between functional requirements, quality requirements and constraints?
- What is the problem with the term non-functional requirement?
- Why is progress in the three dimensions of requirements engineering not linear?
- What are the benefits of requirements documentation?
- What are the differences between the Lastenheft and the Pflichtenheft?
- What are requirements sources and how can we elicit requirements from them?
- Why is conflict identification and resolution important? Why are conflicts common?
- What is the difference between verification and validation?
- What are validation techniques for requirements and how do they relate to the principles of validation?
- What are important attributes for requirements?
- What are advantages and disadvantages of traceability management?
- How is (cyber) security reflected in requirements engineering?

Literature

- | | |
|-------------------------------|--|
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